

SpacECE India Foundation

Smart Learning, Screening & Assessment Platform

Intern Task Allocation Document | 7-Member Team | Confidential

Project Overview

SpacECE (www.spacece.in) is a Pune-based early childhood education non-profit working with children aged 0–8 years in urban slums and rural Maharashtra. This platform is an AI-powered digital learning, screening, and assessment tool that complements SpacECE's on-ground programs (HAALS, UMANG, Parent-Toddler Program).

Children learn through games and activities while the system continuously measures literacy, numeracy, cognition, creativity, and emotional development — generating school readiness reports for parents and teachers.

Tech Stack	Platform	Target Users
React, Vite, Three.js, Tailwind CSS, JavaScript	Web (mobile-responsive PWA)	Children (1–10), Parents, Teachers, Admins
Firebase Auth, Firestore, Storage, Cloud Functions	GitHub + Firebase Hosting	India-context: Hindi/Marathi language support

Individual Intern Task Allocations

Pratiush

Child Learning World & User Journey

Pages to Build & Content Breakdown

► Welcome & Onboarding Page

- Animated welcome screen with SpacECE branding and character mascot
- Age group selection (1–3, 4–6, 7–10) with illustrated age cards
- Child name entry with emoji keyboard for young users
- Form data saved to Firestore child_profiles collection

► Avatar Selection & Customization Page

- Gallery of 12+ diverse avatar characters
- Accessory slots: hat, outfit, badge, background
- Coin counter showing earned coins and unlockable items
- Preview panel showing full avatar before saving
- Save avatar changes → update Firestore avatars collection

► Child Dashboard

- Continue Learning card (last incomplete activity)
- Daily Challenge card (randomized activity for the day)
- My Stars counter (total stars earned across all modules)
- My Achievements preview (last 3 badges unlocked)
- Recommended Activities (fed from AI recommendations collection)
- Circular progress ring showing weekly learning goal

► Achievement Page

- Badge gallery: Literacy Star, Math Champion, Creative Mind, Emotion Explorer, etc.
- Trophy shelf with animated display for top achievements
- Certificate section: printable PDFs for parents
- Progress indicators: X/Y badges unlocked per category
- Reward notifications: confetti + sound on new unlock

► Adventure World

- Illustrated adventure map with zones (Literacy Land, Math Mountain, Brain Bay, etc.)
- Treasure hunt interface: click zones to enter learning modules
- Learning journey path showing completed vs locked zones
- Zone completion stamps and XP bar

Unique Features & What More Can Be Explored

- Multilingual support toggle (English / Hindi / Marathi) – important for SpacECE's India context
- Offline mode indicator – many target users may have unstable internet

- Parent PIN lock on child profile switch (for multi-child households)
- Mascot character that reacts to child's activity (idle animations, encouragement messages)
- Dark mode / night mode for bedtime learning
- Printable achievement certificates as styled PDFs
- Explore: Seasonal themes (festivals like Diwali, Holi) that reskin the adventure map

Firestore / Database Collections

- child_profiles (name, age_group, avatar_id, coins, stars, last_active)
- avatars (character, accessories, unlocked_items)
- achievements (badge_id, child_id, date_earned, type)
- rewards (item_id, coin_cost, category, is_unlocked)

Admin Panel Responsibilities

- Add / edit avatar characters and accessories
- Create new badge types and define unlock conditions
- Set daily challenge activity IDs
- Manage reward coin costs

Key Deliverables

- 5 fully responsive React pages
 - Firebase Auth integration for child profiles
 - CRUD on child_profiles, avatars, achievements, rewards
 - Module description + collection schema doc
 - Functional + UI test report
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Harsh

Literacy & Language Learning Universe

Pages to Build & Content Breakdown

► Reading World Page

- Story library with cover thumbnails, difficulty labels, and age tags
- Search and filter: by age group, language, topic
- Featured story carousel at the top
- Story card: title, author, reading level, estimated time

► Story World (Reading Screen)

- Full-screen story reader with illustrated pages
- Text highlighting as audio narration plays (future: TTS integration)
- Comprehension questions after each story
- Story completion screen: stars earned, comprehension score, next story suggestion
- Bookmark and resume reading

► Vocabulary Zone

- Picture matching game: tap the correct image for a spoken/shown word
- Word building: drag letter tiles to form words
- Vocabulary flashcard deck with illustrated meanings
- Score tracker and streak counter per session

► Language Challenges Page

- Alphabet Adventure (Age 4–6): animated letters, trace and match
- Sound Matching: match animal sounds to animals (Age 1–3)
- Object Recognition: tap the named object on screen (Age 1–3)
- Reading Fluency Challenge (Age 7–10): timed reading passages
- Story Comprehension Quiz (Age 7–10): multiple choice + open response

Unique Features & What More Can Be Explored

- Text-to-Speech (TTS) narration using Web Speech API – stories read aloud with page sync
- Phonics mode: highlight each syllable as it is spoken for early readers
- Reading streak system: 7-day streak unlocks bonus story
- Custom story upload for teachers (admin panel)
- Explore: Regional language stories (Marathi folk tales) for cultural relevance
- Difficulty adaptive engine: if child scores >80% twice, system auto-suggests harder story
- Explore: Audio recording for child to read aloud and compare with model pronunciation

Firestore / Database Collections

- stories (title, content_url, age_group, difficulty, language, topic)
- reading_activities (activity_id, type, age_group, content)

- vocabulary_games (game_id, word_list, image_urls)
- language_scores (child_id, activity_id, score, accuracy, time_taken, date)

Admin Panel Responsibilities

- Upload and manage stories (title, age group, language, content)
- Create vocabulary game sets with word-image pairs
- Manage alphabet and phonics activity content
- View reading scores per story

Key Deliverables

- 4 fully responsive React pages
 - Story reader with Firebase Storage integration
 - CRUD on stories, reading_activities, vocabulary_games, language_scores
 - Age-group filtered activity rendering
 - Module doc + screen list + test report
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Pages to Build & Content Breakdown

► Math World Page

- Number activity hub: tap, count, drag number tiles
- Counting games: count animated objects on screen (Age 1–3)
- Number matching: match digit to quantity (Age 4–6)
- Arithmetic challenges: addition, subtraction, multiplication (Age 7–10)
- Score display and star rating after each session

► Puzzle World Page

- Shape matching: drag shapes to correct slots (2D and 3D using Three.js)
- Jigsaw-style puzzle with 4–16 pieces depending on age
- Logic challenges: odd-one-out, sequence completion, mirror shapes
- Timer mode (optional) for advanced players

► Number Adventure Page

- 3D adventure map (Three.js) where completing math activities unlocks new zones
- Interactive number line for subtraction/addition visualization
- Count Toys activity (Age 1–3): tap toys as they appear, count aloud
- Size comparison: drag to order items from smallest to largest

► Logic Island Page

- Pattern games: complete the next item in sequence (colors, shapes, numbers)
- Maze challenges: navigate a character through a maze using directional logic
- Problem solving: word problems displayed as illustrated scenarios
- Multiplication Quest (Age 7–10): timed tables practice with visual arrays

Unique Features & What More Can Be Explored

- Three.js 3D shapes in Puzzle World – children rotate and place 3D blocks
- Animated 3D adventure map for Number Adventure (unique differentiator in EdTech)
- Visual number arrays for multiplication (sets of dots that group into rows)
- Real-time score animation (number counting up on screen after correct answer)
- Explore: Math storytelling mode – numbers embedded in short narratives
- Explore: Abacus simulation using Three.js beads
- Difficulty ladder: Level 1 → 5 per activity, auto-advances on 3 perfect scores

Firestore / Database Collections

- math_games (game_id, type, age_group, question_set, difficulty)
- puzzle_games (puzzle_id, pieces_url, age_group, shape_type)
- logic_games (game_id, pattern_data, maze_layout, age_group)

- numeracy_scores (child_id, game_id, score, time_taken, level, date)

Admin Panel Responsibilities

- Add math activity sets (question, answer, difficulty level)
- Upload puzzle piece images and define slot positions
- Create logic patterns and maze layouts
- View numeracy scores by activity

Key Deliverables

- 4 fully responsive React + Vite pages
 - Three.js integration for 3D puzzle and map
 - Firebase Firestore CRUD for all collections
 - Age-grouped activity rendering
 - Module doc + test report
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Pages to Build & Content Breakdown

► Brain World Page

- Memory Match game: flip cards to find pairs (6–20 card sets by age)
- Sequence Builder: watch a sequence of colors/shapes, then reproduce it
- Pattern Finder: identify the odd-one-out in a grid
- Maze Challenges: navigate to exit, path revealed on correct choice
- Cognitive score, accuracy, and attempt count tracked per session

► Creativity World Page

- Drawing Pad: freehand canvas with color palette, brush size, eraser
- Color Studio: color pre-drawn illustrations (animals, scenes) digitally
- Story Creator: drag and drop characters/scenes to build a 3-panel story
- Save and share drawings (stored in Firebase Storage)
- Gallery of child's saved creations

► Emotion World Page

- Emotion Recognition: show a face, child taps the correct emotion label
- Friendship Stories: short illustrated scenarios, child chooses the kind response
- Kindness Challenges: daily challenge – describe a kind act you did today
- Decision Making Activities: 'What would you do?' illustrated dilemmas
- Emotion check-in at login: child taps how they feel today (emoji scale)

► Story Choice World Page

- Branching narrative: child makes choices that change story outcome
- 3 story branches per narrative, 3 illustrated outcomes
- Emotion-tagged outcomes (happy, surprised, proud, curious)
- Replay to see all endings – encourages exploration
- Reading comprehension question at story end

Unique Features & What More Can Be Explored

- Emotion check-in widget at dashboard entry – tracks emotional state over time
- Drawing Pad saves to Firebase Storage – parent can view child's artwork
- Branching story engine in Story Choice World – unique interactive narrative format
- Social emotional score tracked separately from cognitive score
- Explore: 'Kindness journal' – weekly written or drawn entry about a kind act
- Explore: Parent notification when child flags a negative emotion 3 days in a row
- Explore: Guided breathing activity that activates if child selects 'angry' or 'sad' at check-in

Firebase / Database Collections

- cognitive_games (game_id, type, age_group, content_data, difficulty)
- creativity_activities (activity_id, type, template_url, age_group)
- emotion_activities (activity_id, scenario_text, choices, correct_response, emotion_tag)
- cognitive_scores (child_id, game_id, score, accuracy, attempts, date)

Admin Panel Responsibilities

- Add cognitive game sets (memory, sequence, pattern)
- Upload coloring templates and story scene assets
- Create emotion recognition scenarios and branching stories
- View cognitive and emotional scores per child

Key Deliverables

- 4 fully responsive React pages
 - Drawing canvas with Firebase Storage upload
 - Branching story engine with emotion tagging
 - Firebase CRUD for all 4 collections
 - Module doc + test report
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Pages to Build & Content Breakdown

► My Progress Page (Child-Facing)

- Circular progress rings per domain: Literacy, Numeracy, Cognitive, Creativity, Emotional
- Weekly activity streak calendar
- Star chart: stars earned per day for the last 30 days
- Achievement summary: recent badges unlocked
- Motivational message updated daily based on progress

► Learning Map Page

- Visual map of all learning zones with % completion per zone
- Strength areas highlighted in green, gap areas in amber
- Tap a zone to see activity history and scores within it
- Next recommended activity prominently displayed

► Parent Dashboard

- Child selector (if multiple child profiles)
- Weekly report card: domains, scores, time spent
- Suggested activities for this week (from AI recommendations collection)
- Strength and concern alerts (e.g., 'Reading fluency below age average')
- Download weekly PDF report button

► Teacher Dashboard

- Student list with per-student progress summary
- Class analytics: average scores per domain, bar charts
- Activity assignment tool: assign a specific activity to a student or group
- Learning gap report: students flagged for reading difficulty or numeracy gaps
- Export class data as CSV

► Reports Page

- Historical reports: month-by-month progress for each child
- Comparison: child's score vs age-group average
- Developmental milestones tracker (age 1–3, 4–6, 7–10 benchmarks)
- AI analysis summary: detected patterns, recommendations, flags
- Print / export full developmental report

Unique Features & What More Can Be Explored

- AI gap detection: rule-based engine that flags reading difficulty if accuracy <60% on 3 consecutive sessions
- School readiness score: composite metric based on all 5 domains
- Weekly auto-generated report using Firebase Cloud Functions (scheduled trigger)
- Developmental milestone mapping against NEP 2020 benchmarks

- Explore: Predictive alert – 'Child has not logged in for 5 days' notification to parent
- Explore: Comparative cohort analytics (anonymized class-level trends for teacher)
- Explore: Export reports as beautifully formatted PDFs for printing

Firebase / Database Collections

- assessments (child_id, domain, activity_id, score, accuracy, time, attempts, date)
- progress_tracking (child_id, domain, weekly_score, streak, milestone_flags)
- reports (child_id, report_date, domain_scores, ai_flags, recommendations)
- recommendations (child_id, activity_id, reason, priority, generated_date)
- ai_analysis (child_id, reading_difficulty, numeracy_gap, learning_delay_flag, strength_areas, last_updated)

Admin Panel Responsibilities

- View all assessment data by child and domain
- Manage recommendation rules and thresholds
- Generate manual reports for selected child
- Configure AI flag thresholds (e.g., accuracy cutoff)

Key Deliverables

- 5 fully responsive React pages
 - AI gap detection engine with Firestore writes
 - Cloud Functions for weekly report generation
 - Charts using Recharts or Chart.js
 - Module doc + test report + API documentation
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Vishal

Platform Backend, Firebase Architecture & API Layer

Pages to Build & Content Breakdown

► Firebase Architecture & Security Rules

- Define and implement Firestore security rules for all collections
- Role-based access: child, parent, teacher, admin roles
- Validate data writes: score must be 0–100, age_group must be valid enum
- Authentication rules: unauthenticated users cannot read child data

► CRUD API Layer (Cloud Functions)

- createChildProfile: validates and writes to child_profiles
- updateActivityScore: atomic write to assessments + update progress_tracking
- generateWeeklyReport: scheduled Cloud Function, runs every Sunday midnight
- getRecommendations: reads ai_analysis, returns top 3 activities for child
- flagLearningDelay: triggered when 3 consecutive low scores detected

► Cross-Module Integration

- Coin reward trigger: when any learning module completes, fire Cloud Function to add coins to child profile
- Achievement unlock trigger: when milestone condition met in any domain, write to achievements
- Notification queue: write to notifications collection when parent alert conditions are met

► Admin Panel Backend

- CRUD endpoints for all admin-managed content (stories, activities, avatars)
- Analytics aggregation: compute class averages, domain summaries
- Bulk import: accept JSON upload of activity sets
- Export: generate CSV download of assessment data

Unique Features & What More Can Be Explored

- Atomic batch writes: score update, progress update, and achievement check in a single Firestore transaction
- Offline sync: Firestore offline persistence enabled so children can play without internet and sync when reconnected
- Firestore composite indexes for fast queries (e.g., child_id + domain + date range)
- Rate limiting on Cloud Functions to prevent abuse
- Explore: Firestore aggregation queries for real-time class analytics
- Explore: Firebase App Check to prevent unauthorized API access
- Explore: Scheduled data archiving – move old session data to cold storage after 6 months

Firebase / Database Collections

- All 20+ collections across all 5 modules (owns the master schema)

- notifications (child_id, parent_id, message, type, read_status, created_at)
- audit_logs (action, user_id, timestamp, collection_affected)
- app_config (version, feature_flags, maintenance_mode)

Admin Panel Responsibilities

- Full CRUD access to all collections via admin panel backend
- User management: activate/deactivate accounts
- Analytics dashboard: platform-wide usage stats
- Feature flag management for gradual rollouts

Key Deliverables

- Firebase security rules for all collections
 - 5+ Cloud Functions deployed and tested
 - Cross-module integration (coins, achievements, notifications)
 - Firestore schema documentation with field types and indexes
 - Backend test report + Postman / curl test cases
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Gyanendra

Authentication, Global State, Routing & Shared Components

Pages to Build & Content Breakdown

► Login / Register Page

- Child login: profile select from saved profiles (no password for young children)
- Parent / Teacher login: email + password with Firebase Auth
- Registration: email, role selection (parent / teacher), verify email
- Forgot password flow
- Google OAuth sign-in option

► Profile Management Page

- Parent can create, edit, delete up to 4 child profiles
- Switch between child profiles from parent account
- Teacher profile: add school name, grade, student count
- Account settings: change email, password, notification preferences

► Global Navigation & Layout

- Responsive sidebar/navbar: different menus for child, parent, teacher, admin
- Breadcrumb trail for deep navigation
- Notification bell with unread count
- Bottom navigation bar for mobile (child mode)
- Consistent header with child avatar + coin count

► Shared Component Library

- StarRating component (used across all modules after activities)
- ActivityCard component (used in dashboards and recommendations)
- ProgressRing component (animated SVG, used in progress and dashboard pages)
- ModalDialog component (used for confirmations, rewards, achievements)
- AgeGroupBadge component (color-coded pill for 1-3, 4-6, 7-10)
- LoadingSpinner and ErrorBoundary components

Unique Features & What More Can Be Explored

- Role-based routing: child, parent, teacher, admin each see different pages
- Session persistence: child stays logged in across browser tabs
- Global state with Context API: child profile, coins, current activity synced across all modules
- Protected routes: teacher dashboard requires teacher role verification
- Explore: PWA (Progressive Web App) setup so the platform can be installed on mobile like a native app
- Explore: Lazy loading for each module's routes to reduce initial bundle size
- Explore: Internationalization (i18n) setup for multilingual support

Firebase / Database Collections

- user_accounts (uid, email, role, linked_child_profiles, created_at)
- sessions (uid, login_time, logout_time, device_info)
- notifications (child_id, parent_id, message, read_status, type, created_at)

Admin Panel Responsibilities

- User account management (activate, deactivate, change role)
- View active sessions and login history
- Manage notification templates
- Feature flag UI for toggling platform features

Key Deliverables

- Firebase Auth for all roles (child, parent, teacher, admin)
 - Shared component library used by all 7 team members
 - React Router with protected role-based routes
 - Global state management for child profile and coins
 - Module doc + component Storybook-style reference + test report
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Shared Standards – All Team Members

Code Quality

- Use functional React components with hooks (useState, useEffect, useContext)
- Component files: PascalCase. Utility files: camelCase. CSS modules or Tailwind classes
- No hardcoded Firebase credentials in code – use .env files
- All Firestore writes must include a created_at and updated_at timestamp
- Every page must be tested on both desktop (1280px) and mobile (375px) breakpoints

GitHub Workflow

What Makes This Platform Unique

- India-first context: Marathi/Hindi support, Anganwadi and ICDS alignment, rural-friendly offline mode
- NEP 2020 aligned: developmental milestones mapped to national education policy benchmarks
- School Readiness Score: single composite metric covering all 5 developmental domains
- AI-powered gap detection: not just a game app – it is a developmental screening tool
- Three.js 3D world: adventure map and puzzle world go beyond standard 2D EdTech apps
- Branching narrative stories: social-emotional learning through choice-based storytelling
- Parent capacity building: parents receive actionable weekly reports, not just scores
- SpacECE integration: platform mirrors SpacECE's actual on-ground programs for continuity